

Monad MCP Server

This is an MCP (Model Context Protocol) server that allows Claude Desktop to interact with the Monad Testnet blockchain.

Features

The MCP server provides the following capabilities:

1. **get-mon-balance**: Query the MON token balance for any address on Monad Testnet
2. **get-transaction**: Get details of a transaction on Monad Testnet
3. **get-account-transactions**: Get recent transactions for an address (placeholder implementation)
4. **get-gas-price**: Get current gas price on Monad Testnet
5. **get-contract-events**: Get events emitted by a contract (placeholder implementation)
6. **get-erc20-balance**: Get ERC-20 token balance for an address
7. **get-block-info**: Get information about a specific block
8. **is-contract**: Check if an address is a contract
9. **get-network-stats**: Get current statistics for the Monad Testnet
10. **claim-testnet-tokens**: Request tokens from the Monad Testnet faucet
11. **send-test-tokens**: Send test MON tokens from your own faucet wallet

Setup Instructions

1. Clone this repository:

```
git clone <repository-url>
cd monad-mcp-2
```

2. Install dependencies:

```
npm install
```

3. Configure faucet (optional):

If you want to use the self-hosted faucet functionality, create a `.env` file with:

```
FAUCET_PRIVATE_KEY=your_private_key_without_0x_prefix
```

This allows Claude to send testnet tokens from your own wallet.

4. Build the project:

```
npm run build
```

5. Configure Claude Desktop:

- Open Claude Desktop
- Go to Settings > Developer
- Open `claude_desktop_config.json`
- Add the following configuration:

```
{
  "mcpServers": {
    "monad-mcp-2": {
      "command": "node",
      "args": [
        "/path/to/monad-mcp-2/build/index.js"
      ]
    }
  }
}
```

- Replace `/path/to/monad-mcp-2` with the absolute path to this directory
- Save the file and restart Claude Desktop

Usage Examples

Here are some examples of how to use this MCP server with Claude Desktop:

Checking MON Balance

```
Can you check the MON balance for address 0x1234...?
```

Getting Transaction Details

```
Show me the details for transaction 0xabcd...
```

Checking Gas Price

```
What's the current gas price on Monad Testnet?
```

Getting Block Information

Tell me about block 12345 on Monad Testnet

Claiming Testnet Tokens

I need some test MON tokens. My address is 0x1234...

Checking if an Address is a Contract

Is 0x1234... a contract address?

Getting ERC-20 Token Balance

Check the balance of the token at 0xTOKEN... for the wallet 0xWALLET...

Development

Adding New Tools

To add a new tool to the MCP server:

1. Add its name to the `capabilities` array in `src/index.ts`
2. Define the tool using `server.tool()`
3. Rebuild with `npm run build`

Debugging

If you encounter issues, check the console output for error messages. The server logs detailed information about each request and any errors that occur.

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